21 January 2025

Ancillary Services in Germany: Present, Future and the Role of Battery Storage Systems

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Abstract

The growing integration of renewable energy sources poses new challenges to the existing power system infrastructure. These challenges include reducing power system inertia, resulting in faster frequency dynamics and reduced stability margins. Additionally, the limited capacity of power electronics-based inverter units reduces the sources of reactive power generation, jeopardizing voltage stability. To address these issues, system operators are increasingly required to procure ancillary services to ensure reliable power system operation. This paper presents a comprehensive literature review of the current state of the ancillary services market in Germany, examining the various procurement schemes currently employed and the economics of services. It also highlights the insufficiency of existing ancillary services and identifies potential services that may be necessary to stabilize Germany's future power systems, particularly as renewable energy sources continue to gain prominence. This review aims to benefit academics, researchers, practitioners, and policymakers by enabling them to make informed decisions and effectively navigate the changing energy landscape in Germany.

Keywords

ancillary services, battery energy storage, fast frequency response (ffr), frequency stability, grid reinforcement, power, energy and industry applications, voltage stability

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***Abstract*—The growing integration of renewable energy sources poses new challenges to the existing power system infrastructure. These challenges include reducing power system inertia, resulting in faster frequency dynamics and reduced stability margins. Additionally, the limited capacity of power electronics-based inverter units reduces the sources of reactive power generation, jeopardizing voltage stability. To address these issues, system operators are increasingly required to procure ancillary services to ensure reliable power system operation. This paper presents a comprehensive literature review of the current state of the ancillary services market in Germany, examining the various procurement schemes currently employed and the economics of services. It also highlights the insufficiency of existing ancillary services and identifies potential services that may be necessary to stabilize Germany's future power systems, particularly as renewable energy sources continue to gain prominence. This review aims to benefit academics, researchers, practitioners, and policymakers by enabling them to make informed decisions and effectively navigate the changing energy landscape in Germany.**

***Index Terms*—Ancillary services, battery energy storage, grid reinforcement, frequency stability, voltage stability.**

I. Introduction

The energy policy of the European Union is focused on achieving climate neutrality by 2050. Germany, in particular, aims to reach this milestone even earlier, targeting 2045 [1]. This commitment drives the ongoing transformation of power systems to shift away from conventional power generation facilities and embrace decentralized renewable energy sources, mainly focusing on wind and solar generation [2]. Renewable energy sources now account for over 55% of total electricity generation in Germany [3]. However, Germany's ambition extends further, aiming to cover at least 80% of gross electricity consumption with renewables by 2030. To achieve this goal, the country has implemented the Renewable Energy Act (EEG) amendment, effective from 1 January 2023 [2]. This regulation promotes increased flexibility, mandating transmission system operators (TSOs) and distribution system operators (DSOs) to procure additional ancillary services to ensure the smooth functioning of the grid and maintain grid stability. The evolving energy landscape is anticipated to increase the demand for ancillary services and necessitate a greater diversity of such services, thereby reshaping the ancillary service markets.

The research work presented in this paper is supported by the Bavarian Research Foundation within the scope of the research project "KI-M-Bat."

Energy storage systems, particularly battery energy storage systems (BESS), have the potential to provide grid flexibility and a range of ancillary services across various time intervals, spanning from milliseconds to hours [4], [5]. Their ability to quickly adjust output and low operating costs make BESS units well-suited for supplying frequency reserves, voltage control, and enabling energy arbitrage [5]–[7]. The decreasing costs of battery technologies further enhance their potential in future power systems. Projections indicate that global cost reductions of 75% by 2050 are expected in the best case [6]. Furthermore, the coming fleet of second-life batteries can also be deployed for providing grid ancillary services [8].

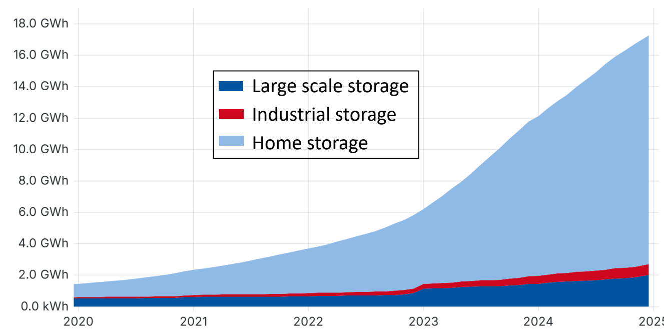

Fig. 1: Battery storage status in Germany as of October 2023 [9]

Fig1 illustrates the current battery storage capacity in Germany, with home storage units contributing 15 GWh, while large-scale storage (LSS) projects account for 2 GWh, and industrial storage adding up to 0.39 GWh. The projected storage demand for LSS in Germany is expected to reach 104 GWh by 2030, primarily to support grid stabilization efforts [10]. To meet this increasing demand and stimulate investment in LSS projects, Germany has implemented a special tariff scheme that provides a 20-year exemption from grid fees for transmission-connected LSS projects commissioned between 2011 and 2026 [11]. Furthermore, on 17 October 2023, the German Energy Industry Act (EnWG) underwent significant amendments (§ 118, subsection 6), extending the grid fee exemption to include LSS installations commissioned prior to 2029 [12]. This particular amendment can further bolster investments in LSS projects.

Approximately two-thirds of the LSS capacity in Germany is dedicated to providing frequency ancillary services [10].

Table. I provides comprehensive information about various LSS projects in Germany and the services they offer. Notably, the primary services observed in these projects are frequency support. However, each country's demand for frequency control reserves remains nearly constant, leading to market saturation and declining reserve capacity prices. This situation poses a discouraging sign for new investments in the sector [13], [14]. As future grid operations require a broader range of ancillary products to effectively manage the intermittency associated with renewable energy generation, regulatory bodies shall identify these future ancillary services and establish a profitable market structure that encourages BESS participation in grid stabilization.

A. *Paper contribution*

Numerous studies have been conducted on the requirements for new ancillary services and potential market structures; however, none have specifically focused on the unique requirements of the German power system. This research addresses this gap by making two significant contributions:

(i) Providing a detailed overview of the German electricity market, the status quo of the current ancillary services, procurement mechanisms, and the economics involved.
(ii) Examining additional ancillary service products needed to accommodate the increasing share of renewable energy sources in the German power system, along with implementation examples from different countries.

The remainder of the paper is structured as follows: Section (II) provides an overview of the German electricity market, briefly explaining the wholesale and ancillary service markets. Building upon that, Section (III) discusses the current status of ancillary services in Germany, including both frequency and non-frequency services. The procurement mechanism and prevailing prices of the ancillary services are presented, with comments on the feasibility of BESS units for market participation. Section (IV) highlights the inadequacy of current ancillary products in coping with the increasing share of inverter-based generation and recommends potential future ancillary products to maintain grid stability. Section (V) focuses on the potential frequency and non-frequency ancillary services that may be required in German power systems in the near future. The section also provides examples of similar services from different countries. Section (VI) highlights the congestion problem in Germany and showcases the opportunities for grid reinforcement deferral through BESS at both the transmission and distribution levels. Finally, Section (VII) offers concluding remarks and recommendations regarding the future of ancillary services in Germany.

II. German electricity market: definitions, legal framework, and current status

The EnWG reform of 1998 incorporated the liberalization of the electricity sector to decouple the physical flow (transmission and distribution) and the commercial flow (trading and sales) [15]. The liberalization of the energy sector introduced various new players into the market. In 1999, the balancing groups were introduced into the energy market. The balancing groups take care of the commercial flow of electricity, while the physical power transfer is managed by the TSOs and DSOs. Every consumer or producer of electricity in Germany is signed to a balancing group [15]. The utility providers (or the retail suppliers) have an agreement with the balancing group to supply electricity to its consumers. The balancing group system takes electricity trading into account. At the wholesale level, the traded electricity is transferred from the seller's balancing group to the buyer's balancing group.

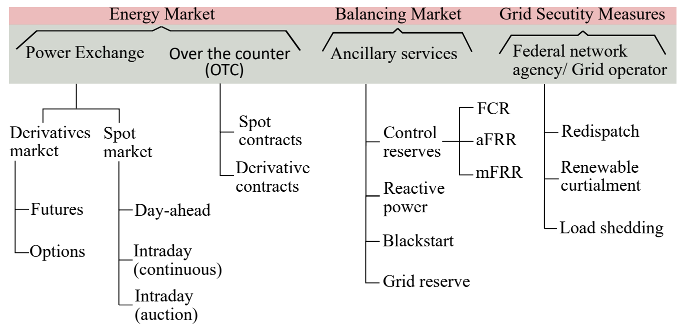

Fig. 2: Summary of electricity market design in Germany

According to the market report from the German Federal Cartel Office (Bundeskartellamt) [29], Germany had 890 DSOs in 2021. Their core activity is to maintain and monitor distribution grids. In addition, they assign consumption points to balancing groups and invoice grid usage fees. The TSOs only invoice customers who are directly connected to the transmission grid, including distribution grids. Through this mechanism, all customers connected to the distribution grid pay a share of transmission costs. In addition to maintaining the transmission grid, it is the responsibility of TSOs to maintain the grid frequency at a stable 50 Hz. To achieve this, TSOs are required to procure capacity reserves to balance any load and generation intermittency. The balancing groups offer the capacity reserves through auctions through various market platforms. Fig. 2 summarizes the German electricity market and Table. II presents the current market share of major German electricity companies at different operational levels.

A. *Wholesale electricity market*

Electrical energy can be traded at either power exchanges or over-the-counter (OTC), i.e. with bilateral contracts. There are different trading windows. The trading activities in Germany primarily take place on three exchanges: the European Energy Exchange (EEX) located in Leipzig, the EPEX SPOT in Paris, and the Energy Exchange Austria (EXAA) in Vienna [28]. The long-term contracts up to 7 years in the future are traded on the derivative market through EEX platform. For the intraday and day-ahead contract purchases, EPEX and EXAA platforms are utilized, respectively. For the OTC trade, both long-term and short-term products exist.

B. *Ancillary services definition*

The EU Directive 2019/944 [30]:

TABLE I: List of selected large-scale battery projects for providing different grid services in Germany

Project	Service	Technical Details	Economic Details (if available)
Energy Storage Bordesholm [16]	Ancillary services - FCR, Voltage support, Black start, Islanding	10 MW and 15 MWh capacity, Saving of 12,000 tonnes CO2 equivalent per year, Grid area of Tennet	€10 million investment, €1 million annual turnover
Battery Storage Jardelund [17]	Bulk energy services - Electric energy time shift (arbitrage), Reduce wind energy curtailment	48 MW and 50 MWh capacity, Consists of 10,000 Li-ion battery modules, Owned by Mitsubishi and Eneco Holding	€30 million investment
Energy Storage Alt-Daber [18]	FCR participation, Balancing of volatile PV generation	2 MW Storage and 2 MWh capacity, Owner & Operator: Upside Group	€1.3 million investment
BigBattery Lausitz [19]	Ancillary services - FCR, Electric energy time shift (arbitrage), Grid stabilization	50 MW and 53 MWh capacity, 8,840 battery modules	€25 million investment
Schwerin Battery Park [20]	Ancillary services - FCR, Black start, Islanding, Electric energy time shift (Arbitrage)	10 MW and 15 MWh capacity, Funding from the federal government – R&D, private funding	€10 million investment
Battery 2nd Life Hamburg [21]	Ancillary services - FCR	2 MW and 2.8 MWh capacity, 2600 modules from more than 100 cars, Second life battery project, Owned by Vattenfall, supported by BMW and Bosch	N.A
Pre-life Hannover [22]	Ancillary services - FCR	15 MW and 15 MWh capacity, 3000 battery modules, Second life battery project, Owned by Daimler and Stadtwerke Hannover	N.A
Netzbooster Kupferzell [23]	Grid congestion management/ Immediate upgrade deferral	250 MW and 250 MWh capacity, Owned by TransnetBW, To be finished by 2025	€200 million investment
Netzbooster Audorf South and Ottenhofen [7]	Grid congestion management/ Immediate upgrade deferral	250 MW and 250 MWh capacity, Owned by Tennet GmbH, To be finished by 2034	€200 million investment
M5BAT Aachen [24]	Research project: Battery life optimization for Control reserve trading, Arbitrage	5 MW and 5 MWh capacity, Li-ion battery, lead acid and Na-NiCl2 batteries, Project at E.ON Energy Centre, Supported by: SMA, Exide, Beta Motion and RWTH Aachen	€6.5 million grant from the German Federal Ministry for Economic Affairs
Wind+Storage project Brandenburg Uckermark [25]	Grid flexibility, Avoid curtailment	Germany's first Wind + Storage project, 3 MW and 3 MWh capacity	N.A
Six projects: Herne, Lünen, Bexbach, Fenne, Weiher, and Walsum Storage System North Rhine-Westphalia [26]	Ancillary services - FCR	90 MW and 45 MWh capacity in total, 15 MW and 7.5 MWh capacity each, Li-ion batteries, Owned by STEAG GmbH	€100 million investment

Article. 2 (48), defines the **"ancillary service"** as *"a service necessary for the operation of a transmission or distribution system, including balancing and non-frequency ancillary services, but not including congestion management"*.

Article. 2 (49), defines the **"non-frequency ancillary service"** as *"a service used by a transmission system operator or distribution system operator for steady state voltage control, fast reactive current injections, the inertia for local grid stability, short-circuit current, black start capability and island operation capability"*.

§ 13e of EnWG mandates TSOs to ensure frequency stability within their respective control areas [31]. Similarly, § 12h of EnWG [32] mandates both TSOs and DSOs to procure non-frequency system services for voltage stability within their respective operation areas. To compensate for active power deficits, TSOs procure reserve capacity through the balancing markets, a separate component from electricity markets. The capacity reserve is established via a competitive tendering process ensuring transparency and non-discrimination. Additionally, TSOs conduct an annual system analysis to determine the necessary 'reserve capacity' for future grid stabilization through re-dispatch. The German Federal Network Agency (Bundesnetzagentur, BNetzA) reviews the system analysis findings and publishes an annual report on the requirement for reserve generation capacity [33]. Control reserve and grid reserve collectively constitute ancillary services [34].

C. Grid security measures from Grid Operators/ Federal Network Agency

In an electrical power system, generation and demand must be balanced at every time, and it is the responsibility of TSO to ensure frequency stability, and TSOs divide their responsibilities into smaller units known as balancing groups. Each balancing group ensures that the generation, consumption, and traded energy within that specific zone are in equilibrium. To facilitate this balance, each balancing group is required

TABLE II: German electricity sector market share in 2023 [27] [28]

Sector	Market leaders	Market share	Total number of providers
Transmission	Amprion, TransnetBW, TenneT, 50Hertz	100% combined	4
Distribution	EnBW, E.ON, RWE, Vattenfall,	The big 4 companies own and operate a significant portion (25.8%) of the distribution grids	Approx. 890 DSOs, with about 700 of them owned by *Stadtwerke*
Generation	RWE, LEAG EnBW, E.ON, Vattenfall, Others	Share by generation: RWE (25.3%), LEAG (14.9%), EnBW (9.9%), E.ON (9.6%), Vattenfall (5.6%), Others (34.7%)	Over 1000 producers
Retail suppliers	EnBW E.ON RWE Vattenfall	They have a 62% share in the total retail market	Over 1000 retailers

to provide the TSO with a 15-minute resolution forecast detailing all anticipated activities for the following day [15]. After receiving forecasts from all balancing groups, the TSO conducts a power-flow study to identify potential grid bottlenecks. A grid bottleneck occurs when certain transmission lines lack the necessary capacity to accommodate the traded energy flow. In such cases, adjustments need to be made to the power allocation through a process known as “re-dispatch." This involves reducing the generation output of units located upstream of the bottleneck and activating units located downstream. The owners of the down-regulated units are financially compensated for their participation in the re-dispatch measures and the units upstream are compensated for the power curtailment.

III. Ancillary Service Market: Status Quo

A. *Frequency support ancillary products*

In power systems, uncertainty is always associated with forecasted generation and demand, as well as technical disturbances (e.g., power plant outages, line faults, cyber-attacks). Therefore, the TSOs are required to maintain active power reserves available in real-time, and in case of a shortfall, the TSOs must take remedial actions. In Germany, the TSOs procure three different types of reserves (FCR, aFRR, and mFRR), each reacting at a different time scale, as illustrated in Fig. 3. Due to the different reaction times, specific generation units are considered for each of the reserves.

1) Frequency containment reserve (FCR): The FCR is equivalent to the primary control reserve. It is the initial control action and is self-activated within a synchronous area. Following a disturbance, the FCR should be fully available within 30 seconds and capable of supporting the grid with 100% capacity for at least 15 minutes. The FCR is currently

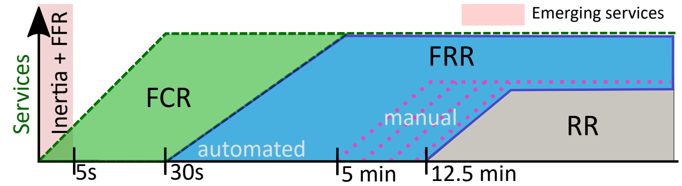

Fig. 3: Balancing reserves (reserves are bidirectional)

being supplied as part of a joint tender, known as FCR co-operation, by the TSOs of Germany, Austria, the Netherlands, Belgium, France, and Switzerland and is accessed through a common market. In 2019, BNetzA introduced amendments to the FCR market to make it more accessible for small stakeholders [35]. These changes included the introduction of the "15-minute criteria," which requires a continuous and uninterrupted supply of the agreed amount of FCR for at least 15 minutes during an "alert state". Previously, the minimum activation time was 30 minutes [36].

An alert state is defined as a frequency deviation of $|\Delta f| \geq$ 50 mHz for 15 minutes, $|\Delta f| \geq 100$ mHz for 5 minutes, or instantaneous $|\Delta f| \geq 200$ mHz [37]. If the frequency deviation is less than 200 mHz, the system should provide a partial delivery for a proportionately longer period. It is crucial that the supply begins immediately when the deviation occurs and continues for at least 15 minutes after the alert state is triggered. The FCR product has the following characteristics:

(i) The FCR demand is determined on a European level, according to ENTSO-E guidelines.
(ii) FCR is a symmetric product, i.e., upward and downward reserves are procured together. For example, 1 MW FCR represents 1 MW of regulation both upward and downward [37].
(iii) The auction for FCR takes place daily, offering six four-hour products for the following delivery day. An auction closes 1 day before the delivery at 8am [38].
(iv) Divisible and indivisible bids are allowed by TSOs. Indivisible bids have a maximum bid size of 25 MW in all participating countries. The minimum bid size is 1 MW. Capacity can be offered in steps of 1 MW up to the prequalified power [39].
(v) The synchronous area Central Europe (CE) consists of several load-frequency control (LFC) Blocks. Each LFC Block consists of one or more LFC Areas. The Monitoring Areas, LFC Areas, and LFC Blocks for Germany are summarized in Table. III.
(vi) The required FCR volume for LFC areas is annually calculated on the basis of each LFC area’s power generation and demand. The FCR volumes can also be traded among the LFC areas within an LFC block [38].
(vii) The ramping rate of the FCR product is TSO specific and is determined at the time of pre-qualification [40].
(viii) As of October 2023, Germany’s estimated demand for FCR reserves is 573 MW [41], with BESS systems constituting two-thirds of the German FCR market [42].

2) Frequency Restoration Reserves (FRR): The FRR reserve is further classified into two categories: automatic Fre-

TABLE III: List of Monitoring Areas, LFC Areas, and LFC Blocks in Germany, adapted from [39]

Country	TSO (full name)	TSO (short name)	Monitoring Area	LFC AREA	LFC Block
Germany	TransnetBW GmbH	TransnetBW	TNG	TNG	TNG+TTG+AMP+50HZT+EN+CREOS
	50Hertz Transmission GmbH	50Hertz	50HZT	50HZT	
	TenneT TSO GmbH	TenneT GER	TTG	TTG+EN	
	Amprion GmbH	Amprion	AMP	AMP+CREOS	

quency Restoration Reserves (aFRR) and manual Frequency Restoration Reserves (mFRR), which act as secondary and tertiary frequency reserves, respectively. The deployment of aFRR should reach 100% capacity within 5 minutes. In the case of grid imbalances lasting longer than 5 minutes, mFRR is activated to support aFRR. The trading of aFRR reserves takes place through the ENTSO-E enabled Platform for the International Coordination of Automated Frequency Restoration and Stable System Operation (PICASSO) [43], while mFRR reserves are traded through the Manually Activated Reserves Initiative (MARI) [44]. Due to the longer delivery duration, small-scale BESS units are unsuitable for FRR services. Instead, synchronous generators, flexible industrial loads, virtual power plants, and pump storage systems are typically utilized [45]. Following are the characteristics of FRR products [41]:

(i) aFRR and mFRR are procured as separate products through PICASSO and MARI market platforms.
(ii) aFRR and mFRR are asymmetrical products, and a separate auction is held to procure positive and negative products.
(iii) The products shall have a minimum bid size of 1 MW.
(iv) Daily auctions offer six four-hour aFRR and mFRR products for the next delivery day.
(v) aFRR auction closes at 9 a.m., one day before delivery, and mFRR closes at 10 a.m., one day before delivery.

3) Replacement reserves (RR): RR is the last market available control reserve and is deployed manually. Typically activated from 15 min after the imbalance occurrence and can stay up to hours until the normal operation is restored. The RR products can be brought from the balancing service providers (such as generators or demand facilities) or exchanged between the TSOs through the Trans European Replacement Reserve Exchange market platform (TERRE) [46]. Currently, Germany does not procure any RR product [47]. Following are the characteristics of RR product [47]:

(i) The common platform TERRE has 24 trade phases that are carried out throughout the day.
(ii) The ramping period of a product can range from 0 to 30 minutes. However, the total activation time for the product cannot exceed 30 minutes.
(iii) The smallest volume increment in which bids can be submitted is 1 MW. The delivery duration of the product can be 15, 30, or 60 minutes.
(iv) The minimum size of a bid is 1 MW. In the case of a divisible bid, there is no maximum limit, but only technical limits are constraints. In the case of indivisible bids, specific national rules are implemented [48].
(v) Only six TSOs at present, namely ČEPS, REE, REN, RTE, Terna, and Swissgrid, are connected through the TERRE platform [49].

B. Non-frequency ancillary products

Non-frequency ancillary services encompass a range of services used by TSOs and DSOs to ensure grid stability [32]. While these services are primarily provided by generators, they can also be offered by demand facilities, network operations and equipment, and energy storage facilities [50]. Various research studies at the European level [51]–[53], are dedicated to the establishment of a market for non-frequency ancillary products. These projects aim to enhance TSO-DSO coordination and establish local non-frequency ancillary service markets based on optimal power flow [54]. However, in Germany, there is currently no common market mechanism in place for procuring non-frequency ancillary services [55]. These services are acquired through bilateral agreements, with pricing determined through private negotiations and not disclosed to the public, making it non-transparent [56].

1) Voltage Regulation: Status Quo: According to ENTOS-E [57]: *"Voltage Control refers to the control actions designed to maintain the set voltage level or the set value of Reactive Power"*. The following are the key points regarding voltage control in Germany.

(i) In Germany, reactive power procurement is primarily dedicated to fulfilling only steady-state requirements [58].
(ii) At the transmission level, voltage stability is the responsibility of TSOs, who are required to maintain their reactive power compensation units [15].
(iii) Additionally, network operators can enter into bilateral contracts with individual generation plants to procure reactive power for system stability [56].
(iv) Reactive power can be sourced through various means [58], including: (i) Capacitor banks (ii) Synchronous compensators (iii) High Voltage Direct Current (HVDC) links (iv) On-load tap changing transformers (OLTC) on Extra High Voltage (EHV) or High Voltage (HV) transformer sides (v) Energy storage units (vi) Renewable Energy Sources (vii) Conventional power plants.
(v) According to the Demand Connection Code [59], the relevant TSO has the authority to specify the reactive power range for transmission-connected demand facilities, including large industries and DSOs.
(vi) Currently, there is no established regulatory framework governing the settlement rules between TSOs and DSOs (also transmission-connected demand facilities and industries) regarding the pricing of reactive power exchange. It typically depends on individual bilateral contracts [58].

(vii) The code also mandates that the power factor of power imported by DSOs (also transmission-connected demand facilities and industries) must not fall below 0.9 [59].

(viii) To regulate reactive power, DSOs in Germany have the flexibility to choose between different fixed power factor control methods, including cos(ϕ), cos(ϕ(P)), and Q(V) droop control, for local voltage regulation [52].

(ix) In the event of voltage-related issues at the DSO level, corrective measures are undertaken by the TSOs after exchanging necessary information [52].

(x) TSO-connected DSOs and demand facilities are penalized if they fail to adhere to the specified power factor regulations as decided in the bilateral contracts [52].

(xi) Power-generating modules in Germany are categorized based on voltage levels and installed generation capacity. Detailed specifications of these four generator types in Germany and their respective reactive power capabilities are summarized in Table IV and Table V, respectively.

TABLE IV: Categories of power-generating modules in Germany [60]

Generator	Requirements	Active power capacity	Voltage level
Type A	Capabilities to withstand wide-scale events. Limited automated response and control.	$0.8\,\mathrm{kW} \leq P < 135\,\mathrm{kW}$	$< 110\,\mathrm{kV}$
Type B	Automated dynamic response. System operator control.	$135\,\mathrm{kW} \leq P < 36\,\mathrm{MW}$	$< 110\,\mathrm{kV}$
Type C	Stable and controllable dynamic response. Covering all operational network states.	$36\,\mathrm{MW} \leq P < 45\,\mathrm{MW}$	$< 110\,\mathrm{kV}$
Type D	Wide-scale network operation and stability. Balancing services.	$P \geq 45\,\mathrm{MW}$	$\geq 110\,\mathrm{kV}$

TABLE V: Reactive power capability for the generating modules [61]

Module Type	Type B	Type C, D
Synchronous power modules	Specified by the system operator	$U - Q/P_{\max}$ profile with a maximum range of 0.95 for $Q/P_{\max}$
Power park generation modules	Specified by the system operator	$U - Q/P_{\max}$ profile with a maximum range of 0.75 for $Q/P_{\max}$
Offshore power park modules	$U - Q/P_{\max}$ profile	$U - Q/P_{\max}$ profile with a maximum range of 0.75 for $Q/P_{\max}$

C. *Economics of Ancillary Products in Germany*

According to § 12h EnWG [32]; BNetzA is responsible for evaluating the potential for "transparent, non-discriminatory, and market-based" procurement of certain services in the energy sector. If market-based procurement is found to be economically inefficient for a particular service, BNetzA must define an exception and specify an alternative procurement method. Based on these guidelines, the frequency control reserves are auctioned via a market platform. In the case of non-frequency ancillary services, only voltage regulation services and the black start services are deemed to be procured through market procedure. However, currently, no market mechanism is in place for them [56].

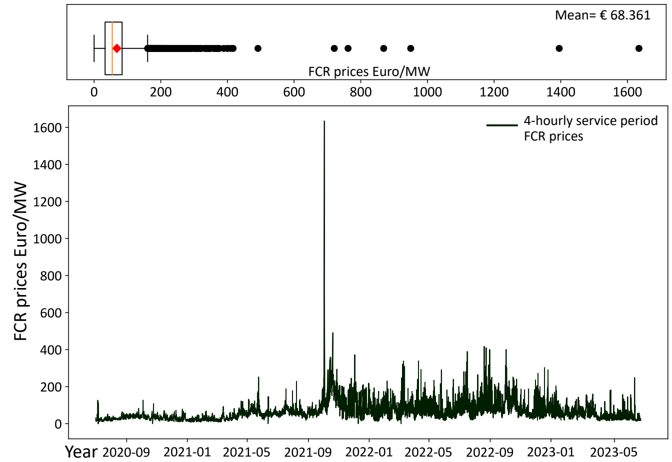

Fig. 4: FCR price development in Germany from the beginning of 4-hr service period (since July 2020), data retrieved from [41]

The FCR prices have decreased by more than 50% from 2015 to 2021 [13]. Furthermore, the price trends of FCR reserves, as shown in Fig. 4, have remained considerably low for the last three years, with a mean price of €68.36 since July 2020 [41]. The low prices can be attributed to falling battery prices and market saturation due to the limited FCR volume of 573 MW, causing the FCR market to become less profitable over the years [13], [14], [62]. During the third quarter of 2021, there was a notable surge in FCR prices, which can be linked to the Russia-Ukraine conflict and the uncertainty surrounding the import of Russian gas. Fig. 5 and Fig. 6 illustrate the progression of aFRR and mFRR average prices in Germany for the year 2022, which is considerably low [63]. Research from [64] has indicated that the prevailing aFRR prices in Germany do not offer an economic case for battery storage participation in the market [41]. Additionally, the extended delivery time for mFRR excludes the BESS units from market participation.

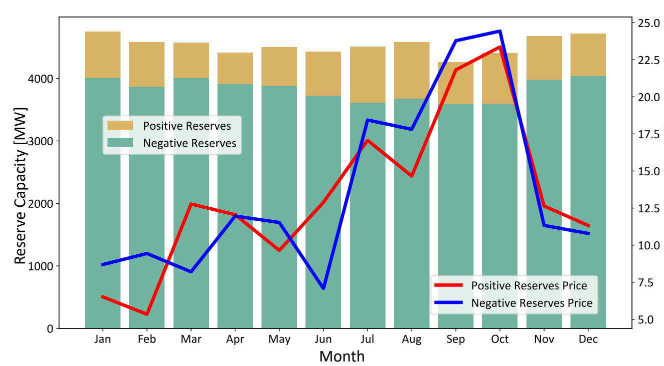

Fig. 5: Average aFRR volume and prices in Germany for the year 2022 [41]

The prices for reactive power are not uniform and can vary significantly due to different bilateral contracts between TSOs and generators/consumers. The cost to customers for reactive power demand can increase to a maximum of 9.20 €/MVArh [57]. Similarly, power plants are compensated by the TSOs for supplying reactive power, but the specific prices are not publicly disclosed and are determined through individual bilateral

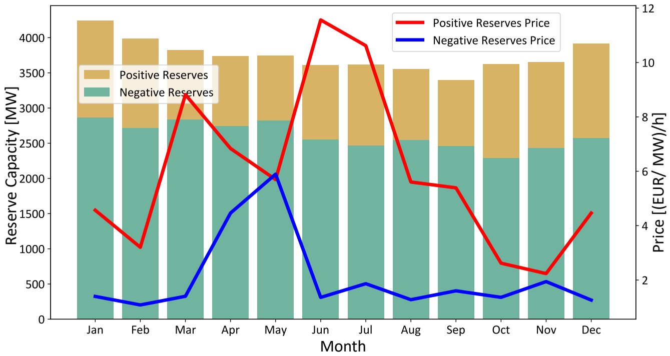

Fig. 6: Average mFRR volume and prices in Germany for the year 2022 [41]

contracts between the parties involved [57].

IV. Inadequacy of Present Control Reserves

The 2023 report from German TSO Amprion on the current status of frequency reserves [65], along with the analysis conducted by the Energy Economics Institute at the University of Cologne (EWI) [66], have emphasized the potential future vulnerabilities to the German power system associated with low levels of inertia. Additionally, studies conducted by European TSOs [67]–[69] have raised concerns about the dimensioning criteria of FCR reserves in Continental Europe, which are based on a 3000 MW power outage assumption [70]. This assumption originates from the 2012 ad-hoc report by the European Network of Transmission System Operators for Electricity (ENTSO-E) [70]. Given the significant increase in the share of renewable energy sources in Europe since 2012, reassessing and updating the dimensioning criteria for FCR reserves is crucial. Fig. 7 illustrates the trend for power system inertia (H) in Continental Europe for the year 2030 [71].

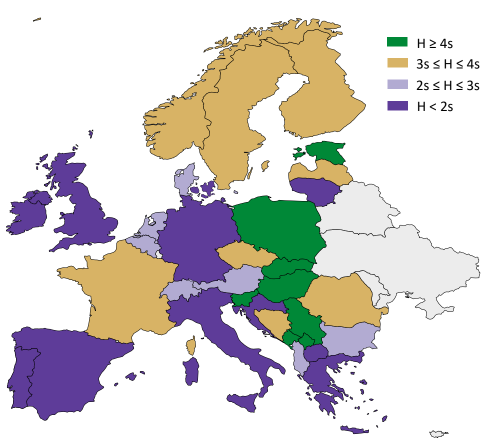

Fig. 7: Expected inertia contribution of countries for the year 2030 [71]

Reduced power system inertia leads to high Rate of Change of Frequency (RoCoF) values. The current FCR reserves cannot address the issue of high RoCoF values due to their slower response time compared to the inertial response [72]. The delay in response can lead to unintentional tripping of lines and generators. For instance, a significant grid disruption occurred in early 2021 in Europe following a sudden 0.25 Hz frequency drop, resulting in the separation of the European interconnected system [69]. A similar frequency event on 10 January 2019 revealed that the grid's capacity imbalance exceeded the primary control reserve capacity of 2.6 GW at the time [73]. Another grid event occurred on 14 December 2018, where an incorrect forecast for photovoltaic feed-in led to a grid imbalance in Germany [73]. Germany's contribution to instantaneous (inertia) reserves is notably low, making it vulnerable to withstand any spilt event, as highlighted by the studies from German TSOs [74], [75]. Therefore, bridging the gap between fault occurrences and the activation of primary frequency reserves is crucial to ensure power system stability.

Voltage stability is also facing significant challenges as renewable energy penetration increases. Research findings from project MIGRATE [76] suggest that the existing voltage control measures alone will no longer be adequate in the future. In Germany, projections suggest a potential reactive power deficit of 38.1 GVar to 74.3 GVar by 2035 [77]. This deficit arises from the inability of inverter-based generation to provide the same level of reactive power as synchronous generators [78]. Addressing the reactive power requirements of wind and solar power plants can be achieved through additional inverters or a STATCOM device. However, this approach requires oversizing inverters, leading to substantial investment costs.

Furthermore, the integration of inverter-based generators results in reduced short-circuit currents due to semiconductor limitations [78]. During transient events, the doubly fed induction generator turbines behave as Squirrel Cage Induction Generators, consuming reactive power and reducing the system's voltage stability limit [78]. The 2021 market report by the German TSO, Amprion [79], has also raised concerns about the diminishing sources of reactive power support in the German power grid and the growing occurrence of rapid voltage fluctuations. To ensure voltage stability, a diverse range of voltage ancillary services will be required in the near future, covering various time scales from milliseconds to hours. The anticipated issues stemming from a substantial increase in power electronics-based renewable generation in the power system can be summarized as follows:

(i) **Frequency stability concerns**: The decrease in inertia resulting from fewer synchronous generators in the power system may lead to frequency instability, particularly in the event of a system split in Central European system.
(ii) **Voltage stability**: Renewable generators often have limited capability for voltage and reactive power regulation, which can lead to local voltage instability, especially at low voltage levels due to dispersed PV generation.
(iii) **Reduced rotor angle stability**: The presence of fewer synchronous generators and faster power system dynamics can negatively impact rotor angle stability.
(iv) **Power oscillations and damping issues**: With the gradual phasing out of synchronous generators, the power system stabilizers may also become less effective, potentially resulting in power oscillations and reduced damping.
(v) **Harmonic and power quality**: Inverter PWM switching, interactions between power converters and passive com-

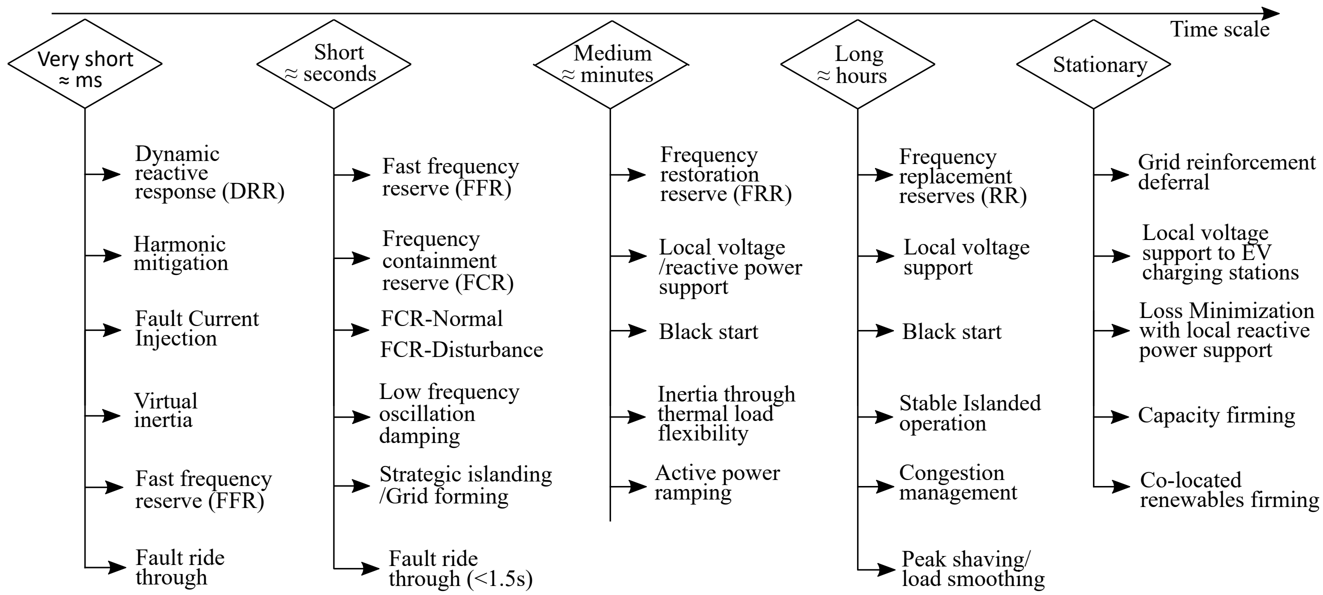

Fig. 8: Overview of potential grid ancillary services in Germany anticipated for the near future

ponents in the AC network, leading to potential resonance problems and harmonic magnification.

V. Ancillary products: Future requirements

The preceding sections have presented the current status of grid ancillary services in Germany and have highlighted the inadequacies of these services in supporting the existing German power system infrastructure as it prepares for an increased share of renewables in the near future. This section shifts the focus to the anticipated needs and emerging solutions for enhancing the resilience of German power systems. Fig. 8 shows the complete list of potential ancillary services. The services are classified based on the required delivery time scales. The section begins with examining future frequency support ancillary products, offering a rationale for why these services may be needed in Germany, and showcasing implementation examples of similar services from various countries. Subsequently, the future non-frequency ancillary products are explained in detail.

A. Frequency support ancillary products

1) Fast Frequency Reserves (FFR): FFR refers to the rapid adjustment of active power, involving both generation and load, within a short time frame (typically 2 seconds or less) [80]. The significance of FFR services in ensuring stability in low inertia power systems has been emphasized in the report by the North American Electric Reliability Corporation (NERC) [81]. ENTSO-E report on future inertia [82], have estimated the average power system inertia for continental Europe as illustrated in Fig. 7. To address the challenges posed by low inertia phases, the Nordic countries have already implemented FFR as an ancillary product [83]. Battery storage systems and demand side response are usually utilized for the FFR services [84], [85]. Similar to the FFR product of the Nordic grid, Great Britain has also introduced a new suite of dynamic response services [86], that includes three dynamic ancillary products operating at different time resolutions: (i) Dynamic Containment, (ii) Dynamic Moderation, and (iii) Dynamic Regulation. The idea behind dynamic products is to counter reduced system inertia and help restore the frequency faster. Table. VII summarizes the FFR products from different TSOs around the world, along with the product specifications.The IEEE standard P2800-2022 [87] has provided the classification and technical specifications for FFR services. However, the implementation of FFR services can be challenging because during an event of fault, there can exist different local frequencies and different Rates of Change of Frequency (RoCoF) at different buses [72]. Therefore, this may lead to false triggering of services.

2) Virtual inertia services: In the future, the responsibility of providing virtual inertia will be assumed by inverters [88]. However, the market and incentives for virtual inertia in the ancillary services market are currently lacking [89]. The German network development plan for 2021-2035 has introduced inertia requirement guidelines for the TSOs [29]. In response to these guidelines, the German TSOs are already working towards establishing a separate market and incentives for virtual inertia to ensure its effective integration and utilization within power systems [90]. Virtual inertia and FFR are two different ancillary products, according to the Australian Energy Market Operator (AEMO) [85]:

"Inertia provides an inherent response to slow the RoCoF, but cannot act to restore power system frequency. $\neq$ FFR can inject active power to correct the imbalance and restore system frequency, but does not inherently slow RoCoF in the same manner."

The Australian Energy Market Commission (AEMC) has requested the establishment of an inertia spot market to ensure grid stability [91]. The release of the procurement mechanism draft is expected by February 2024. Similarly, Great Britain is also moving towards introducing inertia procurement through bilateral contract mechanisms. During the COVID-19 pandemic, the electricity system in Great Britain encountered difficulties due to the significant presence of renewable energy sources coupled with reduced electrical demand. This situation compromised the stability of the grid due to a lack of inertia. To ensure stability, the British TSO had to secure substantial capacity reserves. However, this excessive compensation resulted in a three-fold increase in the cost of procuring ancillary services during the months of May to July 2020 compared to the same period in 2019 [92]. To address the high costs of ancillary services, a real-time inertia monitoring framework has now been established, along with the introduction of bilateral virtual inertia contracts [93]. These contracts, valued at £323 million, are scheduled to begin in April 2024. The virtual inertia is allocated between synchronous condensers and grid-forming converters, with the latter representing a pioneering large-scale implementation [93], [94].

3) Frequency containment reserve enhancement products: According to a joint paper by the four European TSOs: Swissgrid, Amprion, Stattnet, and Red Eléctrica de España [68], the establishment of energy markets and extensive renewable energy generation could potentially result in insufficient volumes of FCR. Therefore, it may be advantageous to consider the re-dimensioning of current FCR reserves and the introduction of new FCR products on top of it. For example, the Nordic TSOs realized the insufficiency of the FCR volumes in the Nordic synchronous area [95]. Driven by the need to handle stochastic imbalances and sudden disturbances arising from the decreasing inertia conditions [95], the Nordic TSOs have introduced two separate FCR products: FCR-N, designed for normal operating conditions with a volume of 600 MW, and FCR-D, intended for large disturbances with a volume of 1450 MW. The overall FCR volume was increased from 1450 MW to approximately 2050 MW (FCR-N + FCR-D). Adopting a similar approach in Continental Europe can lead to an increase in the FCR dimension for stability enhancement and the expansion of the market volume. Moreover, this approach creates opportunities for small-scale battery storage systems to participate in the market. By leveraging the dynamic parameter tuning, these BESS units can efficiently provide both FCR-N and FCR-D products [96], opening more revenue streams.

4) Damping power system oscillations: The introduction of HVDC interconnections, offshore wind farms, and power electronic devices, combined with weak grid conditions, is leading to an increased occurrence of power oscillations globally. Some of these incidents are summarized in Table VI. For example, integrating offshore wind power plants has resulted in subsynchronous oscillations at 4 Hz in the Texas grid and 30 Hz in China [97]. Traditionally, Power System Stabilizers (PSSs) have been used in synchronous generators to mitigate power oscillations [98]. However, with the gradual phase-out of synchronous generators, the responsibility for damping these oscillations is shifting to Inverter-Based Resources (IBRs) [99], [100].

Reports from the MIGRATE project [104] have also highlighted the urgent need to address Low-Frequency Oscillations (LFOs) in the network. The WECC Renewable Energy

Oscillation Source	Location	Frequency (Hz)
LCC-HVDC	Yongfu, Guangxi, China	3
Wind Farm	Texas, USA (ERCOT)	4
Wind Farm	Jilin, Tongyu, China	3-6
Wind Farm	Hornsea, U.K	8.5
MMC-HVDC	Fujian, Xiamen, China	23.6-25.2
MMC-HVDC	Nanhui, Shanghai	20-30
Wind Farm	Xinjiang, China	30
BESS	Moloka, Hawai	18

TABLE VI: Power oscillation events observed around the world [101]–[103].

Modeling Task Force [105] has identified the potential of Battery Energy Storage Systems (BESS) in providing damping ancillary services. Additionally, research [106]–[108] has demonstrated the effectiveness of inverters in damping LFOs and ensuring the stability of microgrids. Currently, there is no established market for electromechanical damping services. However, a report by NREL [109] advocates for introducing battery storage systems to provide damping services in the United States. Similarly, research by [110] highlights the impact of increasing wind penetration on sustained oscillations in the Irish grid and recommends incentivizing generators to offer damping services. In Germany, with plans to nearly double its onshore wind capacity to 115 GWh by 2032 and expand the HVDC corridor network [111], the role of damping services will become even more crucial.

5) Ramping margin services: The ramping margin refers to a power unit's predetermined additional capacity to offer the system operator within a particular time frame and duration [5]. With the growing share of variable renewable energy, the net load curve, commonly called the duck curve, becomes more volatile. This necessitates the availability of reserves that can rapidly respond to the fluctuations in net load. Traditionally, the responsibility of meeting net load ramping requirements has fallen on conventional generators [116]. However, ramping services provided by conventional generators are not recognized as separate ancillary services and are only compensated based on the marginal cost of electricity production. When the ramping procured through energy markets is insufficient to handle the increasingly steep and uncertain ramps caused by VRE, the introduction of additional ramping products becomes necessary. Fig. 9 shows the comparison in the shape of the net load curve from 2016 to 2023 [117], it can be realized that: (i) there are steep changes in net load profile, (ii) there is a sharp bidirectional change in the net load (iii) negative net load.

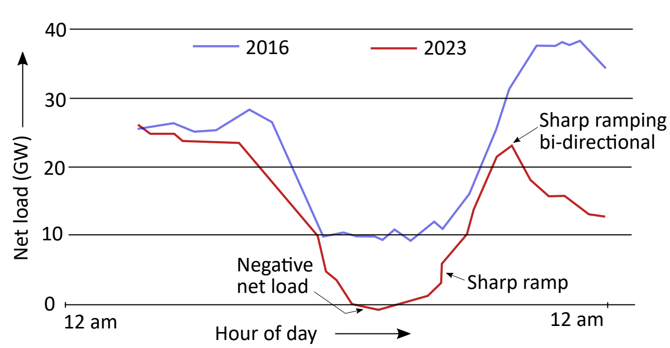

Fig. 9: Germany's net load curve comparison for a specific day (April 10) for the year 2016 and 2023, adapted from [117]

TABLE VII: FFR products and specifications in different countries

Parameters	ONS (Brazil) [112]	CEN (Chile) [112]	EirGrid (Ireland) [52]	ERCOT (Texas) [113]	National Grid (UK) [86]	AEMO (Australia) [114]	Nordic TSOs [115]
Response time	<1 second	<1 second	<2 second	0.5 s (f<59.7 Hz) 0.25 s (f<59.85 Hz)	<0.5 second <0.5 second <2 second	<1 second	1.3 s (f <49.7 Hz) 1.0 s (f <49.6 Hz) 0.7 s (f <49.5 Hz)
Product duration	>5 second	>5 minutes	>8 second	>1 hour >15 minutes	>15 minutes >30 minutes >60 minutes	>6 second	>5 second

Based on the analysis of Germany's net load curve, it is imperative to introduce an independent ramping margin product within the German electricity market. Currently, in Germany, the provision of ramping services is handled by the aFRR and mFRR reserves, with the TSO defining the ramping rate during the pre-qualification process [53], [40]. Recognizing this necessity, several system operators in the USA, including CAISO (2015), MISO (2011), and EirGrid, have taken steps to enhance their energy and reserves markets by incorporating three different ramping products [53], [118].

6) Grid forming capability: Findings from project MIGRATE [76] and study from German TSO Amprion [75] provide evidence that future system splits, with a renewable energy share of 60 to 70%, cannot be effectively managed using the currently implemented grid-following inverters in the German grid. In response to this challenge, the German TSO Amprion emphasizes the urgent need for integrating grid-forming inverters into the German power systems [75]. A joint paper released by the four German TSOs further emphasizes the necessity of deploying grid-forming inverters to ensure stable grid operation, especially with a high share of converter-based generation ($>60\%$) [74]. In addition, the European Union–funded project OSMOSE [119] recommends including grid-forming capability requirements for all new transmission-connected batteries in the European grid.

The Energy Systems Integration Group (ESIG) identifies BESS as "low hanging fruits" due to the significant capacity already installed in existing power grids and easy retrofitting to grid forming control [120]. In a similar line, NREL's recent report [4] recommends the installation of grid-forming technology in new projects and retrofitting old BESS units with grid-forming controls. In Great Britain, the National Grid ESO [121] suggests offering grid-forming capabilities as a paid service (and not mandatory) to encourage greater adoption of grid-forming inverters in future power systems [94]. Considering the concerns raised by German TSOs and the examples of grid-forming inverter technology and matured grid connection specifications from other countries, Germany shall consider deploying new renewable projects and large-scale BESS units with grid-forming control as a proactive measure to ensure stable grid operations in the near future.

B. Non-frequency ancillary products

1) Dynamic Reactive Response (DRR): In the initial phase of fault or sudden change in renewable generation, it is required to deliver a reactive current response for voltage dips in excess of 30% that would achieve at least a reactive power in MVar of 31% of the registered capacity at nominal voltage. The reactive current response shall be supplied with a rise time no greater than 40 ms and a settling time less than 300 ms. BESS presents an excellent potential for providing dynamic reactive power response [5], [122]. The 2021 market report from the German TSO, Amprion, has highlighted the immediate need for dynamic reactive power services in German power systems to encounter the frequent incidents of rapid voltage fluctuations [79]. Fig. 10 shows the fluctuations in transmission voltages due to high renewable intermittency and the inability of existing voltage support measures to ensure voltage stability. Great Britain has been a trailblazer in introducing DRR ancillary services through market mechanisms. Individual DERs and virtual power plants can participate in the market to provide DRR services [123]. Similarly, EirGrid, the Irish TSO, has also included DRR as an ancillary service [118], [124].

2) Voltage harmonics mitigation: Power electronics-based nonlinear loads within the distribution system have become a significant issue for power systems operators, leading to concerns regarding power quality. These loads can introduce voltage distortion, adversely impacting the sensitive loads and interfering with communication systems. Traditional methods for mitigating harmonics involve passive shunt filters that absorb specific harmonics within a defined frequency range. However, these filters are not effective under dynamic operating conditions of a microgrid. An alternative solution is using active filters, which leverage power electronics technologies to produce specific current components that cancel out the harmonics caused by nonlinear devices. It is feasible to provide active filter operation as an ancillary service [125]. For the harmonic mitigation market to work, DSOs and TSOs must adopt a more flexible approach, and system operators should enable the implementation of adaptable grid codes specifically at the grid locations where these services are required.

3) Fault ride through (FRT): Traditionally, power electronics-based generation units were not designed to provide fault ride-through support and would disconnect from the grid during voltage sags or faults. TSOs anticipate a decrease in short-circuit power as renewable energy penetration grows. Consequently, there will be a greater occurrence of voltage dips resulting from short-circuit events, which will affect a larger portion of power generation connected through power electronics. This situation increases the risk of under-voltage protection tripping for such

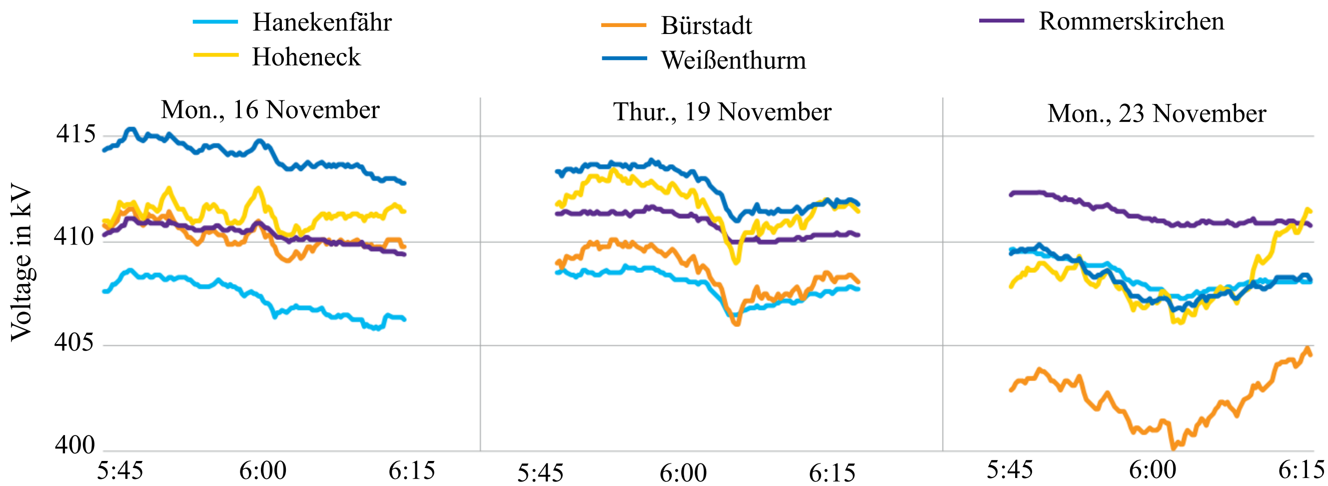

Fig. 10: Voltage fluctuations at different places in German transmission network (control area Amprion) due to high renewable intermittency and limited reactive power support (year 2021) [79]

generation sources [104]. To avoid those problems, these inverter-based units are expected to continue delivering power to the host grid during faults or voltage sags, ensuring uninterrupted power supply to local microgrid loads and offering reactive power support to the grid. This change in perspective highlights how FRT can serve as an additional ancillary product at both the TSO and DSO level [126].

4) Fast Post Fault Active Power Recovery (FPFAPR): Following a transmission fault, if a significant number of generators fail to recover their rated active power output, it can lead to a substantial power imbalance, resulting in severe frequency instability. To ensure system stability, power plants must restore at least 90% of their pre-fault active power output within 250 ms after the voltage level recovers to at least 90% of its pre-fault value, following the clearance of any fault disturbance within 900 ms. Additionally, generators must remain connected to the system for 15 minutes after the fault occurrence. Centrally dispatched synchronous generators, including coal, CHP, biomass, hydro, pumped hydro units, energy storage, synchronous compensators, large-scale battery units, and wind turbines can all provide this service. EirGrid, the Irish TSO, has included FPFAPR as an ancillary service since 2014 [124]. Similarly, in Great Britain, the dynamic response suite has incorporated an identical service for post-fault current injection service scheme [86].

5) Black Start Capability: The black start capability refers to initiating electricity generation without relying on an external power supply from the electricity grid. In order to provide black start services in Germany, the power plant shall be connected to the transmission grid [58]. Currently, conventional plants primarily provide the black start capability [127]. There are 174 black-start capable plants in Germany with a minimum rated output of 10 MW [128]. According to Kyon Energy GmbH, the estimated market size for black start capacities is €7.4 million. Additionally, in 2019, German TSOs contracted a capacity of 4.6 GW for black start reserves [128]. Traditionally, black start capability has been procured through bilateral negotiations and contracts with conventional plants, such as pumped-storage, natural gas-fired, and occasionally coal-fired power plants. These black start services are primarily compensated for their availability, enabling them to recover specific investments, operational costs, and maintenance expenses. In some cases, TSOs may provide warming costs to ensure the units remain available rather than being shut down. According to § 12h (9) of EnWG [32], under certain circumstances, TSOs can require storage operators to maintain the black start capability of their plants and be compensated accordingly. The report from the German Energy Agency (DENA) [127] recommends reinforcing plants with black start capabilities and introducing an ancillary service market for procurement. Procuring through the ancillary service market can offer large-scale battery storage facilities the opportunity to participate [128]. ERCOT, the Texas system operator, established an annual competitive black start market in 2009 [129].

6) Peak shaving: Peak shaving is a strategy industrial and commercial power consumers employ to smooth out the spikes in their electricity usage. By implementing peak shaving techniques, these consumers aim to achieve a more consistent and manageable energy consumption pattern [130]. The primary objective is to minimize the occurrence of high peaks in electricity demand, which can strain the power grid and lead to inefficiencies or even blackouts. The EU Electricity Market Revision Survey [131] has suggested introducing a peak shaving ancillary service that system operators can purchase to encourage demand reduction and energy shifting during peak times. Likewise, the European TSOs have also requested the introduction of peak shaving as a flexibility product [132], [133]. This product can be auctioned with a capacity payment and activated during peak load with an energy payment [131]. China is leading the way forward in establishing the market for non-frequency ancillary services. Various provinces in China have already introduced peak shaving ancillary products through a spot market mechanism [134], [135]. The provinces of Fujian, Gansu, Shandong, Jiangsu, Xinjiang, and Guangxi have carried out peak shaving ancillary products through the spot market scheme [136].

7) Arbitrage services: Although arbitrage services are not included in ancillary services, stacking other ancillary services with wholesale arbitrage market participation can improve the profitability of BESS investments [137], [138] [139]. The fundamental principle of power price arbitrage is to exploit the variability of electricity prices in the trading market for financial gain. This strategy involves purchasing electricity when prices are low and storing it for later sale when prices increase, resulting in a profit. The profitability of power price arbitrage depends on the difference between the lowest and highest points during a given period. The frequent occurrence of negative electricity prices in Germany can support the profitability of the arbitrage business model. Negative electricity prices offer an opportunity to purchase electricity at extremely low or even negative costs (being paid for storing the power) [140]. In the German market, the introduction of 15-minute trade blocks in the continuous intraday market has enhanced the flexibility for balancing agents to adjust their positions and take advantage of arbitrage opportunities [15].

VI. Investment Deferral with batteries

The increasing share of renewable power sources (RES) and the additional delays in Germany's planned grid expansion projects pose challenges for TSOs in their day-to-day grid operation and operational planning [141]. TSOs frequently need immediate remedial actions, such as re-dispatching, to alleviate grid congestion. Re-dispatch measures are independent of the balancing services. All generation or storage facilities that a TSO obliges to adjust their active power for re-dispatch are remunerated based on reported costs [142]. However, in situations with high RES feed-ins, the availability of suitable power plants for re-dispatching is often insufficient [52].

Consequently, curtailments of renewable energy feed-ins are implemented as emergency measures in accordance with EnWG [143]. The curtailment process is implemented sequentially, starting with the reduction in RES connected to the transmission grid, followed by RES connected to the distribution grid. This process involves the TSO requesting the DSO to curtail the electricity feed-in. Unlike re-dispatch measures, a curtailment is an unbalanced approach that necessitates additional balancing measures, including balancing power. As a result, compensational payments are made for curtailing RES, along with payments for the required balancing energy. This situation results in double payments [52], [27].

A. Congestion relief and re-dispatch

In Germany, the issue of curtailments in the electricity grid has resulted in significant compensation claims, amounting to approximately €761.2 million in 2020. A new framework, "re-dispatch 2.0," was implemented in response to these losses on 1 October 2021. This updated regulatory framework allows smaller generating plants with a capacity of 100 kW or more to participate in the re-dispatch model. This change aimed to involve both TSOs and DSOs in mitigating congestion and maintaining the grid's stability.

According to a report from Deutsche Bank [144], the cost of managing grid congestion through various measures amounted to €771 million in the second quarter of 2023, significantly higher than the €389 million recorded in the same period of 2022. One notable limitation of the current re-dispatch framework is the exclusion of storage systems, electric cars, and demand-side potential as viable options for congestion management [145]. These resources possess flexibility that could aid in alleviating congestion and stabilizing the grid. However, due to the cost-based approach employed in re-dispatching, which primarily focuses on evaluating fuel and labor costs, grid operators face challenges in accurately assessing the dispatch costs associated with storage resources. Consequently, there is a hesitancy to utilize storage systems due to uncertainties in cost forecasts [145]. Furthermore, according to Amprion, huge power imbalances are expected between the north and south of Germany by the year 2030 [146], as illustrated in Fig. 11.

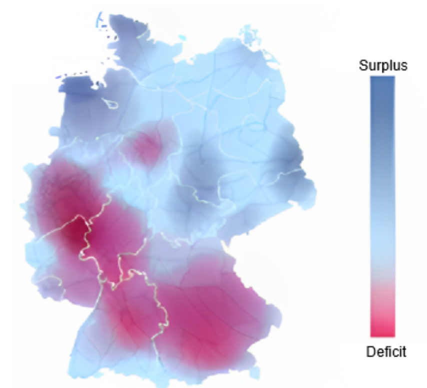

Fig. 11: Expected power imbalance in Germany in the year 2030 [146]

The launch of the "re-dispatch 3.0" framework [147] in the near future may alleviate the congestion problems by allowing producers and consumers with nominal ratings below 100 kW to participate in a re-dispatch market. This inclusion will enable the involvement of electric cars, home battery storage, and heat pumps in re-dispatch measures. Although the market design for re-dispatch 3.0 is not yet established, TSOs are working on potential concepts through research projects and field trials [148]. Nevertheless, introducing the "re-dispatch 3.0" framework can potentially create significant economic opportunities for battery asset owners [55].

B. Transmission grid upgrade deferral: Grid boosters

A total grid reinforcement cost of €198 billion is estimated for the whole transmission network in Germany by 2037 [144]. Large-scale battery storage systems have also gained recognition as an effective solution for managing congestion in the power grid and avoiding the re-dispatch costs at the transmission level. The success of this approach is evident from the installation of three large batteries by the French Transmission System Operator (RTE) at the sub-transmission grid level [149]. In Germany, the unbundling regulations for electricity storage facilities have been in practice since 31 December 2020, prohibiting system operators from owning the storage facilities. However, in some cases, the network operators may own energy storage facilities if these plants qualify as fully integrated network components [150]. The German TSOs have collaboratively proposed the implementation of network boosters as pilot projects in their draft for the 2030 network development plan [7]. Amprion TSO is currently constructing a decentralized network booster in the Bayrisch-Schwaben area, which will utilize batteries with a capacity of 250 MW. This project is scheduled to become operational in 2025 [7]. Similarly, Tennet TSO GmbH is developing two pilot projects in Audorf South and Ottenhofen, each with a capacity of 250 MW, to be connected to the grid before 2034 [7]. TransnetBW TSO has also planned a 250 MW battery project near Kupferzell and Höpfingen and is scheduled to commence operations by 2025 [23].

C. Distribution grid upgrade deferral

In the European Union, a significant portion of low-voltage power lines, approximately 35-40%, were between 20 to 40 years old in 2020, while 25-35% exceeded 40 years in age, according to a report from the International Energy Agency (IEA) [151]. This aging infrastructure presents a pressing issue for Germany's distribution grids. A report by Deutsche Bank [144] underscores the necessity of comprehensive grid reinforcement to ensure reliable operations. This reinforcement is estimated to cost €47 billion by 2037. Similarly, a study from the German Ministry of Economics and Technology [152] suggests that an annual investment ranging from €4.0 to €9.6 billion until 2032 will be required to strengthen the German distribution network.

Traditionally, system operators have addressed congestion issues by investing in new infrastructure to increase the network's initial capacity. However, this approach encounters challenges as peak loads occur only for a few hours each day and are often seasonal. Furthermore, the existing voltage control system at the TSO-DSO interface relies on on-load tap changers (OLTCs) on the EHV/HV transformers, and coordination between TSO and DSO is facilitated through phone communication. The situation becomes more complex with the growing penetration of renewable energy sources, as many OLTCs are already operating at their maximum capacity [52]. Placing a storage asset close to the load center can effectively meet electricity demand during peak hours without the need for costly upgrades to the incoming transmission or distribution lines, thus providing a significant economic advantage [153] [154].

According to [155], batteries, even at their current costs, can potentially defer the need for transformer reinforcement at the distribution level by up to 15%. According to [156], a cost-benefit of 45% is possible through obviating the distribution grid reinforcement with flexibility measures mainly provided by BESS. Work from [157] has presented a detailed cost comparison of different reinforcement measures for European distribution grids, highlighting that the strategic placement of battery storage systems provides a promising solution to defer the expensive upgrades of distribution grids below 50% penetration scenario. In summary, by strategically locating storage units, the grid operators can effectively manage peak demand, optimize grid utilization, and minimize the need for costly infrastructure investments.

VII. Conclusion and Recommendations

The paper analyzed Germany's current electricity market structure, focusing on current grid ancillary services, procurement mechanisms, and pricing of services. It also commented on the suitability of BESS units for providing those services. The study also sheds light on the limitations of the existing ancillary products in effectively addressing the challenges posed by integrating renewable energy sources into the grid while suggesting the new set of ancillary services that may be required in the near future. It also commented on the suitability of BESS units for providing those services across various time scales, highlighting their strong potential as contributors. Literature findings indicate that German system operators have acknowledged these challenges and are actively advocating for the inclusion of new ancillary services. This necessitates decisive action from regulatory agencies to collaborate with system operators, identify future ancillary services, and establish profitable procurement mechanisms. The conducted literature review has led to the following recommendations:

1) **Re-dimensioning of control reserves**: The size of frequency control reserves in the Continental European grid shall be re-evaluated and increased to accommodate the renewable intermittency and withstand any split event. Additionally, considering stochastic planning of reserves on a daily basis can help improve their effectiveness and economic efficiency.
2) **Introducing fast frequency products**: To address the challenge of grid inertia, it is suggested to introduce new fast-frequency products or virtual inertia products in the near future. Learning from the implementation challenges from other countries, it is important to establish clear pre-qualification requirements that facilitate easy and profitable participation.
3) **Deploying grid forming inverters**: Grid forming technology is commercially available and field-proven for BESS systems/ hybrid BESS PV systems. It may be beneficial to explore the option of retrofitting existing BESS plants with grid-forming technology. Furthermore, new projects shall already be mandated with grid-forming controls to support future 100% renewable generation-based power systems.
4) **Enhancing TSO-DSO coordination**: Improving the coordination scheme between TSOs and DSOs is essential. The active involvement of DSOs in congestion management through peak shaving services helps alleviate line congestion for TSOs and, most importantly, obviates the need for immediate grid upgrades.

5) **Establishing a regional market**: Establishing regional markets is recommended to efficiently procure non-frequency ancillary services. Insights can be gained by studying the example of Great Britain and China's local market for non-frequency ancillary services. Furthermore, introducing peak shaving products to alleviate grid congestion shall also be considered. China's regional peak-shaving markets can be considered an example of building a market structure.
6) **Leveraging BESS resources**: Germany can enhance its grid stability by utilizing existing BESS resources. However, to facilitate this, a new market framework should be established. This framework should incentivize prosumers to actively participate in the flexibility market by offering home battery storage and EVs. Encouraging a greater number of decentralized smaller prosumers can also help reduce the risk of a single point of grid failure.
7) **Empowering aggregators**: Encouraging aggregators in the ancillary market benefits both prosumers and aggregators. It expands prosumer participation, enhancing grid flexibility, mitigating single points of failure, and increasing aggregator revenue with more participants. It also provides prosumers with a range of aggregator options to choose from, thus creating a win-win situation.
8) **Flexibility in grid codes**: To introduce services such as local voltage support, voltage harmonics mitigation, oscillation damping, and grid forming capability, it is necessary to incorporate flexibility into grid codes. Consequently, TSOs and DSOs must consider revising grid codes at critical bus locations where these advanced functionalities are needed.

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